

### ত্রিকোনোমিতিক সূত্রাবলী

$$\sin(-\theta) = -\sin \theta$$

$$\tan(-\theta) = -\tan \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\operatorname{cosec}(-\theta) = -\operatorname{cosec} \theta$$

$$\cot(-\theta) = -\cot \theta$$

$$\sec(-\theta) = \sec \theta$$

$$1. \sin \theta = \frac{1}{\operatorname{cosec} \theta}$$

$$2. \operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$3. \cos \theta = \frac{1}{\sec \theta}$$

$$4. \sec \theta = \frac{1}{\cos \theta}$$

$$5. \tan \theta = \frac{1}{\cot \theta}$$

$$6. \cot \theta = \frac{1}{\tan \theta}$$

$$7. \sin^2 \theta + \cos^2 \theta = 1$$

$$8. \sec^2 \theta = 1 + \tan^2 \theta$$

$$9. \operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$$

$$10. \sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$11. \sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$12. \cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$13. \cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$14. \sin(A+B) \sin(A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$$

$$15. \cos(A+B) \cos(A-B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$$

$$16. \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$17. \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$18. \cot(A+B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$$

$$19. \cot(A-B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$$

$$20. 2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$21. 2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$22. 2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$23. 2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$24. \sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$25. \sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$$

$$26. \cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$27. \cos C - \cos D = 2 \sin \frac{C+D}{2} \sin \frac{D-C}{2}$$

$$28. \sin 2A = 2 \sin A \cos A$$

$$29. \sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$$

$$30. \cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$31. \cos A = \cos^2 \frac{A}{2} - \sin^2 \frac{A}{2} = 2 \cos^2 \frac{A}{2} - 1 = 1 - 2 \sin^2 \frac{A}{2}$$

$$32. 1 + \cos 2A = 2 \cos^2 A$$

$$33. 1 - \cos 2A = 2 \sin^2 A$$

$$34. 1 + \cos A = 2 \cos^2 \frac{A}{2}$$

$$35. 1 - \cos A = 2 \sin^2 \frac{A}{2}$$

$$36. \sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$$

$$37. \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$38. \cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$39. \sin A = \frac{2 \tan \frac{A}{2}}{1 + \tan^2 \frac{A}{2}}$$

$$40. \tan A = \frac{2 \tan \frac{A}{2}}{1 - \tan^2 \frac{A}{2}}$$

$$41. \cos A = \frac{1 - \tan^2 \frac{A}{2}}{1 + \tan^2 \frac{A}{2}}$$

$$42. \sin 3A = 3 \sin A - 4 \sin^3 A$$

$$43. \cos 3A = 4 \cos^3 A - 3 \cos A$$

$$44. \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

$$45. \sin \theta = \sin \alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$$

$$46. \cos \theta = \cos \alpha \Rightarrow \theta = 2n\pi \pm \alpha$$

$$47. \tan \theta = \tan \alpha \Rightarrow \theta = n\pi + \alpha$$

$$48. \sin \theta = 0 \Rightarrow \theta = n\pi$$

$$49. \tan \theta = 0 \Rightarrow \theta = n\pi$$

$$50. \cos \theta = 0 \Rightarrow \theta = (2n+1) \frac{\pi}{2}$$

$$51. \sin \theta = 1 \Rightarrow \theta = (4n+1) \frac{\pi}{2}$$

$$52. \sin \theta = -1 \Rightarrow \theta = (4n-1) \frac{\pi}{2}$$

$$53. \cos \theta = 1 \Rightarrow \theta = 2n\pi$$

$$54. \cos \theta = -1 \Rightarrow \theta = (2n+1)\pi$$

### ত্রিকোনোমিতিক সমীকরণ

$$45. \sin \theta = \sin \alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$$

$$46. \cos \theta = \cos \alpha \Rightarrow \theta = 2n\pi \pm \alpha$$

$$47. \tan \theta = \tan \alpha \Rightarrow \theta = n\pi + \alpha$$

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$$52. \sin \theta = -1 \Rightarrow \theta = (4n-1) \frac{\pi}{2}$$

$$53. \cos \theta = 1 \Rightarrow \theta = 2n\pi$$

$$54. \cos \theta = -1 \Rightarrow \theta = (2n+1)\pi$$

## বিপরীত ত্রিকোণমিতি

$$\sin \theta = x \Rightarrow \theta = \sin^{-1} x;$$

$$\sin^{-1} x \neq \frac{1}{\sin x}$$

$$55. \sin^{-1} x = \operatorname{cosec}^{-1} \frac{1}{x}$$

$$56. \operatorname{cosec}^{-1} x = \sin^{-1} \frac{1}{x}$$

$$57. \cos^{-1} x = \sec^{-1} \frac{1}{x}$$

$$58. \sec^{-1} x = \cos^{-1} \frac{1}{x}$$

$$59. \tan^{-1} x = \cot^{-1} \frac{1}{x}$$

$$60. \cot^{-1} x = \tan^{-1} \frac{1}{x}$$

$$61. \sin^{-1} x = \cos^{-1} \sqrt{1-x^2} = \tan^{-1} \frac{x}{\sqrt{1-x^2}} = \operatorname{cosec}^{-1} \frac{1}{x} \\ = \sec^{-1} \frac{1}{\sqrt{1-x^2}} = \cot^{-1} \frac{\sqrt{1-x^2}}{x}$$

$$62. \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

$$63. \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$

$$64. \sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$$

$$65. \tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$$

$$66. \tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy}$$

$$67. \tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} \frac{x+y+z-xyz}{1-xy-yz-zx}$$

$$68. \sin^{-1} x + \sin^{-1} y = \sin^{-1} \{x\sqrt{1-y^2} + y\sqrt{1-x^2}\}$$

$$69. \sin^{-1} x - \sin^{-1} y = \sin^{-1} \{x\sqrt{1-y^2} - y\sqrt{1-x^2}\}$$

$$70. \cos^{-1} x + \cos^{-1} y = \cos^{-1} \{xy - \sqrt{1-x^2}\sqrt{1-y^2}\}$$

$$71. \cos^{-1} x - \cos^{-1} y = \cos^{-1} \{xy + \sqrt{1-x^2}\sqrt{1-y^2}\}$$

$$72. 2\tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2} = \cos^{-1} \frac{1-x^2}{1+x^2} = \tan^{-1} \frac{2x}{1-x^2}$$

## ত্রিকোণমিতির সূত্রাবলী

$$73. \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$74. \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$75. \cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

$$76. \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$77. a = b \cos C + c \cos B$$

$$78. b = c \cos A + a \cos C$$

$$79. c = a \cos B + b \cos A$$

$$80. \tan A = \frac{abc}{R} \cdot \frac{1}{b^2 + c^2 - a^2}$$

$$81. \tan B = \frac{abc}{R} \cdot \frac{1}{c^2 + a^2 - b^2}$$

$$82. \tan C = \frac{abc}{R} \cdot \frac{1}{a^2 + b^2 - c^2}$$

$$83. \sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$$

$$84. \sin \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}$$

$$85. \sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$$

$$86. \cos \frac{A}{2} = \sqrt{\frac{s(s-b)}{bc}}$$

$$87. \cos \frac{B}{2} = \sqrt{\frac{s(s-c)}{ca}}$$

$$88. \cos \frac{C}{2} = \sqrt{\frac{s(s-a)}{ab}}$$

$$89. \tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$$

$$90. \tan \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{s(s-b)}}$$

$$91. \tan \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

$$92. \tan \frac{A}{2} = \frac{(s-b)(s-c)}{\Delta}$$

$$93. \tan \frac{B}{2} = \frac{(s-c)(s-a)}{\Delta}$$

$$94. \tan \frac{C}{2} = \frac{(s-a)(s-b)}{\Delta}$$

$$95. \cot \frac{A}{2} = \frac{s(s-a)}{\Delta}$$

$$96. \cot \frac{B}{2} = \frac{s(s-b)}{\Delta}$$

$$97. \cot \frac{C}{2} = \frac{s(s-c)}{\Delta}$$



## क्यालकुलस

$$01. \frac{d}{dx}(x^n) = nx^{n-1}$$

$$02. \frac{d}{dx}(x) = 1$$

$$03. \frac{d}{dx}(c) = 0$$

$$04. \frac{d}{dx}(e^{mx}) = me^{mx}$$

$$05. \frac{d}{dx}(e^x) = e^x$$

$$06. \frac{d}{dx}(a^x) = a^x \ln a$$

$$07. \frac{d}{dx}(\ln x) = \frac{1}{x}, x > 0$$

$$08. \frac{d}{dx}(\log_a x) = \frac{1}{x} \log_a e$$

$$09. \frac{d}{dx}(\sin x) = \cos x$$

$$10. \frac{d}{dx}(\cos x) = -\sin x$$

$$11. \frac{d}{dx}(\tan x) = \sec^2 x$$

$$12. \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$13. \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$14. \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

$$15. \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$16. \frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$17. \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$18. \frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

$$19. \frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$20. \frac{d}{dx}(\operatorname{cosec}^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$$

$$21. \frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

$$01. \int x^n dx = \frac{x^{n+1}}{n+1}, n \neq -1$$

$$02. \int 1 dx = \int dx = x$$

$$03. \int e^{mx} dx = \frac{1}{m} e^{mx}$$

$$04. \int e^x dx = e^x$$

$$05. \int a^x dx = \frac{a^x}{\ln a}, a > 0, a \neq 1$$

$$06. \int \frac{1}{x} dx = \ln|x|, x \neq 0$$

$$07. \int \sin x dx = -\cos x$$

$$08. \int \cos x dx = \sin x$$

$$09. \int \sec^2 x dx = \tan x$$

$$10. \int \operatorname{cosec}^2 x dx = -\cot x$$

$$11. \int \sec x \tan x dx = \sec x$$

$$12. \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x$$

$$13. \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)}, n \neq -1$$

$$14. \int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b)$$

$$15. \int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b)$$

$$16. \int \frac{f'(x)}{f(x)} dx = \ln|f(x)|$$

$$17. \int \frac{f'(x)}{\sqrt{f(x)}} dx = 2\sqrt{f(x)}$$

$$18. \int \tan x dx = \ln|\sec x|$$

$$19. \int \cot x dx = \ln|\sin x|$$

$$20. \int \sec x dx = \ln\left|\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right|$$

$$21. \int \sec x dx = \ln|\sec x + \tan x|$$

$$22. \int \frac{1}{\sqrt{x}} dx = 2\sqrt{x}$$

$$23. \int \operatorname{cosec} x \, dx = \ln \left| \tan \frac{x}{2} \right|$$

$$24. \int \frac{1}{a^2 + x^2} \, dx = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$25. \int \frac{1}{1 + x^2} \, dx = \tan^{-1} x$$

$$26. \int \frac{1}{a^2 - x^2} \, dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right|$$

$$27. \int \frac{1}{x^2 - a^2} \, dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right|$$

$$28. \int \frac{1}{\sqrt{a^2 - x^2}} \, dx = \sin^{-1} \frac{x}{a}$$

$$29. \int \frac{1}{\sqrt{1 - x^2}} \, dx = \sin^{-1} x$$

$$30. \int \frac{1}{x \sqrt{x^2 - 1}} \, dx = \sec^{-1} x$$

### কতিপয় ধারা

$$01. (1+x)^n = 1 + \frac{n}{1!}x + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

$$02. (1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$$

$$03. (1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$$

$$04. e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$05. a^x = 1 + x \ln a + \frac{x^2}{2!} (\ln a)^2 + \frac{x^3}{3!} (\ln a)^3 + \dots$$

$$06. \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$07. \ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$$

$$08. \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$09. \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$10. \text{ম্যাকলরিণের উপপাদ্যঃ } f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$